

Recording the Brain at Work: The Visible, the Readable, and the Invisible in Electroencephalography

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The electroencephalogram (EEG), the graphic recording of the electric activity of the human brain, kindled far-reaching speculations about the imminent deciphering of mind and brain in the 1930s. Regardless of the thousands of neurons in the human cortex, recording from a person at rest produced a surprisingly regular line oscillating at 10 per second that disappeared at the moment of mental activity. With ever more groups specializing in electroencephalography, however, the deciphering of mind and brain did not materialize but moved further away in the information produced. In the various approaches employed in EEG research, such as the analysis of the graphic code, the search for pathognomic patterns or the imaging of cognitive processing, visualization guided research as well as theorizing, its productivity continued to keep the epistemological question open.

Keywords electroencephalography, visualization of brain waves, graphic method, information coding, brain theory, machine thinking, visual thinking, epistemology of neuroscience

At the end of the twentieth century, advances in visualizing the brain's activity with functional as well as anatomical precision resulted in the emergence of functional neuroimaging as a new discipline. The clarity and persuasiveness of these new images, which also circulate widely across public media, have fostered the assumption, both among the scientists involved and in the general public, that the neurosciences will be a major field of scientific advancement in the still new century. Recently, a group of prominent neuroscientists, for example, published a “manifesto” declaring the imminent advent of a decisive breakthrough in understanding mind and brain (Elger et al., 2004). Whereas “philosophers have struggled over centuries in vain to achieve even an inch of progress,” the time has finally come, so they postulate, to solve the ultimate riddle of consciousness, intentionality, and language by means of new research tools with scientific rigor (Koch, 2004, p. 229). The current fascination surrounding functional neuroimaging is reminiscent of the attraction another visualization technique held earlier in the twentieth century: electroencephalography, the recording of the brain's intrinsic electric activity from the human head. In both cases, a new method kindled far-reaching speculations, especially the potential of tracking the mind by physiological means. Electroencephalography revealed a direct and strange correlation to mental processing. Simply by looking at the recording,

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scientists were able to determine when a person engaged in a task such as mental arithmetic, how long one concentrated on it and when the task was finished. Back in 1930, a German journalist summarized succinctly the hopes kindled by these new methods (Finkler, 1930): “Today the brain writes in secret code, tomorrow scientists will be able to read neuropsychiatric conditions in it, and the day after tomorrow we will write our first authentic letters in brainscript.” That has not yet happened, and today brains may still show unexpected resistance towards current attempts to reveal their mechanism of action. However, even if this is the case, the historian’s task does not stop with drawing this parallel. For the parallel may yet serve as an ideal starting point for investigating the dynamics and trajectories of research tools in order to arrive at a better understanding of the epistemological function of imaging and visualization in brain research. The history of electroencephalography provides ample material for a comparative analysis of the material culture of an imaging technology (Borck, 2005a).

Electroencephalography, A Belated Success of the Graphic Method

Recording the human brain’s electric activity materialized around 1930 in the form of a graphic chart, showing the fluctuations in electric potential over time. Thus, electroencephalography followed the tradition of the graphic method, so well established in experimental physiology since the second half of the nineteenth century (Brain, 1996). During that period, the method had proven its value for tracking bodily movement, particularly intrinsic oscillatory activity difficult to render by other means such as the pulse, breathing rhythm, or changes in blood volume accompanying every heart beat. In these cases, the method generated visual representations of the physiological process under investigation of a quality never seen before. The various sorts of inscription devices, such as the cardiograph, plethysmograph, sphygmograph, etc., recorded the minute details of physiological processes not only with a higher degree of precision compared to a sensing finger or observing eye, with their form of direct, physiological writing, the recording machines also circumvented the problem of finding an appropriate translation of observational data into human language. The recording machines observed with continuous precision and inscribed the changes immediately as a line on paper, without any need for transforming these observations into language. These were the arguments with which Jules-Etienne Marey, the pioneer and *idéologue* of the graphic method, praised its superiority within the first few lines of the introduction to his influential monograph (Marey, 1878, p. i): “Science has before itself two obstacles which hinder its progress: these are, first, the defectiveness of our senses for the discovery of truths and, second, the inadequacy of language to express or to transmit those that we have acquired.”¹ The registration of physiological processes by self-recording instruments promised unmediated access to the phenomena of life “in their own language” (Marey, p. vii): “The graphic method translates all of these changes in the activity of forces in a striking manner which one could call the language of the phenomena themselves.” Allegedly, visualizations generated by the graphic method produced a specific kind of writing, an inscription in nature’s authentic language. The graphic method furnished an “image of objectivity” (Daston & Galison, 1992).

Particularly convincing had been cases in which the method was employed to record processes not only invisible to the human eye but beyond the scope of the human senses, such as the bioelectric signals accompanying muscle contraction or the heart beat. Following

¹I quote from the English translation available in *Brain*, 1996, pp. 446–464.

up on Augustus Désiré Waller's (1887) initial observation, the physiologist Willem Einthoven, for example, spent his entire career recording the electrocardiogram in its authentic shape (Einthoven, 1895; Borck, 1997). The mechanical objectivity of the automatically inscribed curves revealed the secrets of life as a function of the recording apparatus, mirroring the machine-like operations of the body's organs. How did the many successes with the graphic method in experimental physiology and discoveries such as the ECG impinge upon brain research? Given that the brain's electric irritability had been known since Gustav Fritsch and Eduard Hitzig's (1870) famous experiments and David Ferrier's systematic studies (1876), that the electric nature of nervous activity had been known and observed for much longer, and finally that Richard Caton had reported on brain electricity in 1875, surprisingly little has happened along these lines in brain research. Mary Brazier, in her careful study of the history of brain electricity published already half a century ago, reminded the community of a few clusters of activity in Eastern Europe, where Adolf Beck and Wilhelm W. Pravdic-Neminsky concentrated on recording cerebral potentials in animal experiments at the turn of the century (Brazier, 1961). Neurophysiology's center of gravity, however, lay clearly in investigating the peripheral nervous system. And when the psychiatrist Hans Berger (1929) published the human electroencephalogram (EEG) in 1929, he surprised not only his colleagues but also the community of neurophysiologists at large.² While it had previously seemed pointless to expect a coherent signal to emanate from the myriads of neurons inside the brain, now the clear regularity of the recorded signal posed an obstacle, calling for meaningful interpretation.

Even Berger's initial publication did not kindle much interest among neurophysiologists, who remained reluctant to look into these matters. The turning point came only five years later with the public demonstration of the EEG by Edgar Douglas Adrian in Cambridge in May 1934. In 1932, Adrian had been awarded a Nobel Prize (shared with the much older Charles Scott Sherrington) for his investigation of the neuronal signaling code in the peripheral nervous system. By comparison, the recording of brain waves was much easier than the experiments routinely performed in Adrian's lab in Cambridge. In fact, the experiment turned out to be so simple that its success came as an embarrassment. The physiologist found brain waves, the rhythmic electrical activity of the human head, "almost at once," as he later explained (Adrian, 1971, p. 1A–10). At the next meeting of the Physiological Society, Adrian's head was simply wired up to a sensitive electrocardiograph, which (in the hands of his engineer Bryan Matthews) recorded Adrian's brain activity as a line of ink on paper. The British magazine *Spectator* reported on this remarkable public demonstration (Walter, 1934, p. 479):

Adrian and Matthews recently gave an elegant demonstration of these cortical potentials. [. . .] When the subject's eyes were open the line was irregular, but when his eyes were shut it showed a regular series of large waves occurring at about ten a second. [. . .] Then came the surprise. When the subject shut his eyes and was given a simple problem in mental arithmetic, as long as he was working it out the waves were absent and the line was irregular, as when his eyes were open. When he had solved the problem, the waves reappeared. [. . .] So, with this technique, thought would seem to be a negative sort of thing: a breaking of the synchronized activity of enormous numbers of cells into an individualized working.

²Elsewhere, I have argued that the widespread employment of the graphic method in psychophysiology and Berger's early training in this area are the missing links that explain his discovery of the EEG independent of contemporary neurophysiological research (Borck, 2005b).

The machine revealed several layers of regularity in brain waves. The idle brain generated a surprisingly stable, continuous oscillation of about 10 Hz. This basic rhythm disappeared with sensory stimulation, and thus, the machine accurately recorded sensory processing as a biological phenomenon. Moreover, it also documented mental activity such as solving a problem of calculation. The recording machine accessed the operations of the brain machine at the level of its intrinsic organization, or so it seemed. The writings produced by the machine resonated with the brain's mode of operation, literally reflecting the activity along the neurons' communication system. Thus, electroencephalography revealed electric potentials in the well-structured form of brain waves and it supposed this system of potentials to be a kind of language, a "brainscript" awaiting to be deciphered, as the commentator in the German newspaper had put it at the time. Writing, the writing of the recording machine, so it appeared, was the missing link between mind and brain.

With the idea of brainscript, electroencephalography revived Marey's famous promise—and it pushed it into a new dimension. Brainscript was the language of the brain's neurons, and beyond that, it was also "our" language, the code of human thinking and feeling. Brainscript shared an affinity to natural language with the various graphic recordings that dominated physiology since the second half of the nineteenth century; but brainscript operated at the same time on the symbolical plane, the language side of the nature/culture divide. This hybrid character of the EEG, of being a natural phenomenon full of implicit cultural connotations, guaranteed that it would gain enormous and immediate public attention. It was hoped that the new method would provide the means to decipher the brain at work. The scientists involved in this research were certainly careful not to draw too fantastic or spectacular a conclusion. But the quick dissemination of the new method, the rapid establishment of new groups specializing in electroencephalography across the entire Western world, and the massive funding in this area by research agencies such as the Rockefeller Foundation or the Kaiser Wilhelm Society, mirrored a widespread and deep intellectual investment in electroencephalography. The initial experiments had revealed little more than a correlation, a peculiar withdrawal of the regular brain waves in phase with mental processing, but this was an indisputable and promising trace of psychophysiological interaction. From the very beginning, scientists conceived of brain waves as mediators, anchoring the cultural fabric of human activities in the world of the biological. Reports on a supposed fingerprint-like individuality of brain waves (Travis & Gottlober, 1936), on the EEG of thinking (Kennedy et al., 1948), and on personality profiles by means of EEG patterns (Lindsley, 1944) would rapidly follow. In 1939, Adrian, Walter Cannon, and Tracy Putnam nominated Berger for the Nobel Prize and Ulf Svante von Euler wrote a positive evaluation dossier for the Nobel committee.³ It would have been one of the last Nobel Prizes for a discovery using the graphic method, but due to the beginning of World War II, no prizes were awarded in 1940. Berger, knowing nothing of these developments, committed suicide the following year out of despair and depression.

Adrian, the EEG, and Information Coding in the Nervous System

Before their observation, brain waves did not yet exist as something awaiting discovery. Given the historical importance of Adrian's public demonstration of the EEG, the question arises how and when Adrian started to recognize experimental data produced in his lab as evidence for what would eventually become brain waves. Throughout these years, Adrian

³Archives of the Nobelkomittén, Karolinska Institutet, Stockholm; I am grateful to the Nobel Committee for making this information available to me.

was “reaping the harvest due to the new techniques of electronic amplification” (Adrian, 1971, p. 1A–6). The vacuum tube, mass produced during World War I for military communication, had enabled Adrian to investigate the mode of communication in nerve fibers and to analyze the information transmission across the peripheral nervous system. His American friend Alexander Forbes had familiarized him with vacuum-tube amplification immediately after the end of the war and, on the basis of the new technology, Adrian quickly confirmed experimentally the all-or-none principle his teacher Keith Lucas had postulated before World War I (Adrian & Zotterman, 1926). In his Nobel lecture, he summarized his findings (Adrian, 1965, p. 297): “The nerve fiber is clearly a signaling mechanism of limited scope. It can only transmit a succession of brief explosive waves, and the message can only be varied by changes in the frequency and in the total number of these waves.” Wherever Adrian looked, the nerve fiber turned out to be essentially a kind of telegraphy cable. Without a stimulus nothing was going on; upon stimulation the fiber sent a series of very brief pulses. The only apparent difference was that there were only dots and no equivalents to dashes in nervous communication; instead, neurons seemed to code by the frequency of the dot-like pulses.

The many successes along this research trajectory, the overwhelming uniformity of the data recorded in Adrian’s lab that allowed him to formulate the very principles of nervous communication, turned into an epistemological obstacle as they guided expectancies. In this very precise sense, Adrian observed brain waves belatedly (Adrian & Matthews, 1934). Years before the Cambridge experiment, messy data had started to occur. In 1927, for example, a female graduate student in Adrian’s lab had already recorded data that challenged the general applicability of Adrian’s theory of nervous communication—obviously this was little more but an inconsistency to be overcome by further experiments (Adrian & Matthews, 1927). A few years later, however, a more senior researcher in Adrian’s lab, the Dutch biologist Fredrick Buytendijk, again recorded strange, intrinsic oscillations of the electric potentials that did not obey the principles of the universal communication code (Adrian & Buytendijk, 1931). Eventually, the data amounted to a disturbance that Adrian could no longer ignore and he began to search the literature for similar observations. Thus, he learned of Berger’s report on the electroencephalogram (Adrian, 1971, pp. 1A-9f):

When we turned to Berger’s paper [. . .] we were bound to agree that his findings were of considerable, indeed, of exceptional, interest, and we were greatly surprised that no one, apparently, had tried to repeat them. [. . .] We worked in the basement of the Physiological Laboratory which was reasonably free from electrical disturbance [. . .]. We found Berger’s alpha rhythm almost at once.

And “almost at once” the universal code in the nervous system was no more. Electroencephalography undermined Sherrington’s and Adrian’s concept of higher nervous action as reflex integration. In light of the newly discovered continuous brain waves that were later to be called the alpha rhythm the central nervous system was no longer a central telegraphy office processing incoming messages and outgoing commands, but a strikingly active source of intrinsic activity calling for new interpretative metaphors (Adrian, 1946, p. 4):

The difference is so great [. . .]—not merely a more complicated version of the reflex machinery. What we do find, what Berger demonstrated in man, is the continued electrical pulsation in the nerve cells of the brain, modified by an interacting with the signals which arrive from the sense organs, but not

immediately dependent on them. [. . .] Thus in the brain the effect of an afferent message will be like that of an exhortation to a noisy crowd whereas in the reflex pathways it will be like that of an order to a silent and obedient regiment.

In this description from the mid-1940s, Adrian summarized more than a decade of EEG work. In the metaphor of the “noisy crowd” of constantly communicating neurons, the nervous system now reverberated literally around the concept of intrinsic electric activity. Adrian deliberately chose a comparison that liberates the brain from the dictatorial bonds of reflex physiology. But how can “exhortations to a noisy crowd” generate meaning? One possible conjecture here is that Adrian’s use of this metaphor was his way of expressing the inaccessibility of the mind within the brain. During his work on the EEG, Adrian remained decisively skeptical about its meaning and the psychophysiological significance of brain waves in general. According to Adrian, recording and visualizing the brain’s electric activity revealed surprisingly little; above all it revealed the limits of the very method. One figure, in particular, is instructive for analyzing the visualization strategies employed by Adrian (see Figure 1).

Certainly, Adrian’s main aim was to integrate brain waves, and cerebral potential changes in general, into his theory of electric activity in the nervous system. But his way of integrating brain waves into neurophysiology was not a matter of reducing mind to matter; quite the contrary, his efforts were part of a larger strategy to shield the mind from the destructive consequences of simplistic physiological explanations. This famous figure illustrates the point. Originally published in 1934, the figure still points to the origin of Adrian’s EEG work in his investigations of the nervous signaling in animal ganglions. It is a comparison of the intrinsic electric oscillations recorded from a ganglion with those recorded from a human head. The two recordings look almost identical, regardless of the many differences in the circumstances of each experiment, in the employed recording technologies and in the specificities of the recorded potentials. The striking similarity of the two recordings thus is quite obviously the product of a careful editing process, revealing another function of the image beyond the visualization of electric potentials. “E.D.A.” is certainly Edgar Douglas Adrian. Apparently, it costs no more brain than a water beetle’s ganglion to receive a Nobel Prize. What may seem as an insult to human dignity can also be read as a message to his colleagues to mind the gap and to reflect carefully on the limits

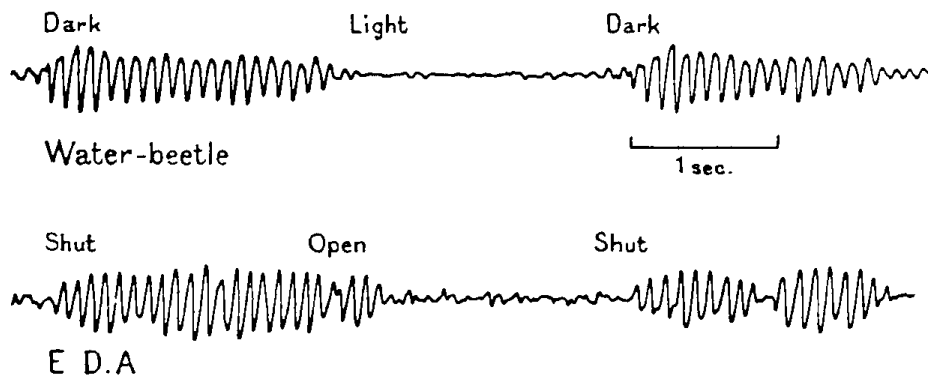


Figure 1. Adrian’s visual assimilation of human brain waves to potential recordings from a ganglion. [Reprinted from E. D. Adrian & B. H. C. Matthews. *The Berger Rhythm: Potential Changes from the Occipital Lobes in Man*. *Brain* 57(4): 373, by permission of Oxford University Press.]

of the method. Seen in this light, Adrian's comparison of himself with the water beetle urged investigators not to rush to a hasty conceptualization of the mind on the basis of the narrow scope of EEG data. More than anything else, the figure illustrates what, according to Adrian, cannot be seen by means of EEG.

Cybernetics and the EEG of Thinking

Adrian was quick to leave the field of EEG research once he had established some basic facts about it—another way to shield the mind from neurophysiological investigation. This may have been a gentlemanly gesture but was hardly an effective strategy to stop the flow of questions and speculations, not even his own. When the theory of cybernetics became available after World War II, Adrian could not resist reflecting in public that cybernetics may offer new approaches to brain waves (Adrian, 1952, p. 296): Using “the theories of the communication engineer and the analogy of the calculating machine to define the functions and powers of the nervous system,” he announced as President of the Royal Society, may “suggest the way in which these functions” are carried out. That was at the end of the year 1951, the same year that had also seen another spectacular EEG experiment. In fact, the result of the experiment was not so spectacular, but the event itself was: it was a kind of brain wave contest between three scientific geniuses, Albert Einstein, John von Neumann, and Norbert Wiener. Above all, this experiment demonstrated the ineradicable persistence of the idea of brainscript.

Unfortunately, only one result has been published (see Figure 2). Neither the EEG of the theory of computation nor that of cybernetics has been made available, only the EEG of the theory of relativity, or to be more precise, the visualization of the electrical activity of Einstein's brain while he was asked to think on the theory of relativity. Roland Barthes famously remarked on this photo (Barthes, 1972, p. 68):

A photograph shows him lying down, his head bristling with electric wires: the waves of his brain are being recorded, while he is requested to ‘think of relativity’. (But for that matter, what does ‘to think of’ mean, exactly?) What this is meant to convey is probably that the seismograms will be all the more violent since ‘relativity’ is an arduous subject. Thought itself is thus represented as an energetic material, the measurable product of a complex (quasi-electrical) apparatus which transforms cerebral substance into power.

Seventeen years of progress in EEG technology and brain wave recording had made surprisingly little difference beyond the technological advances of recording in parallel channels. On all its eight channels, the brain wave recorder reported nothing but a strange withdrawal of the event to be observed. As the *Spectator's* had concluded already in 1934: “So, with this technique, thought would seem to be a negative sort of thing.”

This may have been different in the case of the famous cybernetician, since he had come up with his own theory about brain waves. Wiener was not making light of his contributions to the field when he described his conceptualization simply as the “Rosetta stone” of electroencephalography (Wiener, 1956, p. 289):

[Brain waves] speak a language of their own, but this language is not something that one can observe precisely with the naked eye, by merely looking at the ink records of the electroencephalograph. There is much information contained in these ink records, but it is like the information

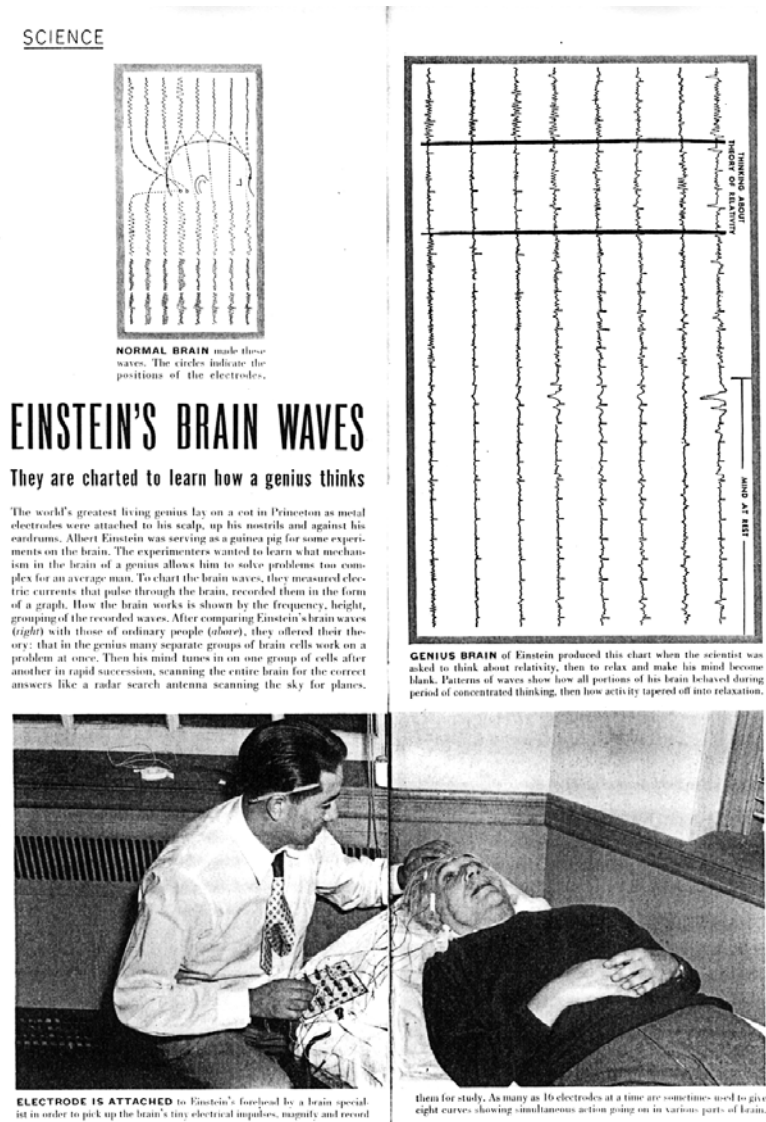


Figure 2. *Life Magazine* reporting on the brain waves of the theory of relativity. [Reprinted from *Life International Magazine*, April 9, 1951, pp. 44–45 with the kind permission of LIFE Branded Products.]

concerning the Egyptian language which we had in the days before the Rosetta Stone. [. . .] When the crude original records of brain waves are transformed by the autocorrelator, we obtain a picture of remarkable clarity and significance, quite unlike the illegible confusion of the crude records which have gone into the machine.

Once again, the EEG wrote in a meaningful language, though in an encrypted form that required analysis (Figure 3). The autocorrelation analysis converted the original recording of brain activity over time into a graph of the distribution of recorded frequencies during a given period. Since experiments suggested a lower correlation of alpha processing in animals than in human beings, Wiener conceived the precision of the alpha rhythm to be an indicator of intelligence. The sharpness of a peak in the autocorrelogram would mirror the accuracy of mental processing. For Wiener, the Rosetta stone of electroencephalography was a sharp peak in the band of alpha frequencies, at exactly 9.05 Hz (Figure 4). The



Figure 3. Norbert Wiener, joined by John Barlow and Walter Rosenblith, watches as the autocorrelogram of his own EEG is computed (Photograph, Research Laboratory of Electronics at MIT, Cambridge, MA, USA, with kind permission).

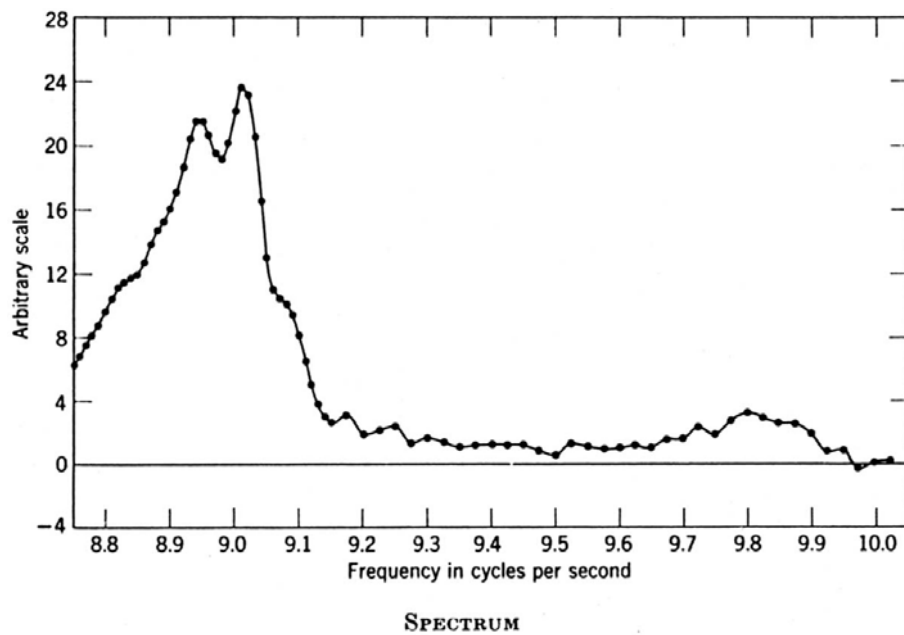


Figure 4. Spectrum of brain wave frequencies, Wiener's example of an autocorrelogram analysis. [Reprinted from Norbert Wiener, *Cybernetics or Control and Communication in the Animal and the Machine*, second edition, Cambridge MA, 1961, by permission of M.I.T. Press.]

comparison with hieroglyphs is hardly a coincidence; like brain waves, hieroglyphs are meaningful symbols *and* mimetic signs. In Egyptian writing, the abstract meaning of the symbolic code is anchored in the iconic reality of the depicted. According to Wiener, brains resembled computers not only in respect of their calculation capabilities but they even used a similar mechanism of data processing, an internal clock for the timing of computation. His theory simply added a calculus to his intuition, but it was hardly a Rosetta stone. First of all, the experiments did not confirm his intuition and no correlation between the stability of the alpha rhythm and intelligence could be observed. But even if such a relation had been established, his hypothesis would still not have been the Rosetta stone, for Wiener's intuition provided a key to deciphering brain waves not as a language but as the mere operating mode of a machine. Rewriting the EEG as an autocorrelogram extinguished any meaning for the purpose of accessing the hardware involved in the coding mechanism.

Writing Code and Thinking with the Eye

Wiener's was not the only approach from cybernetics to brain wave analysis, nor was it the first. By that time, the EEG had already been firmly established as a diagnostic tool. In particular, it had proven indispensable for diagnosing epilepsy and for localizing a brain tumor. Both approaches had started as chance observations. Grey Walter had only incidentally recorded localized, unusually slow potentials from the area surrounding the suspected seat of a brain tumor (Walter, 1936). Also by chance, Frederick Gibbs and William Lennox had recorded the now pathognomic three-per-second spike-wave pattern from a patient with petit mal epilepsy when they teamed up with their Harvard colleagues in electroencephalography (Gibbs, Davis, & Lennox, 1935). In both cases, the meaning of the EEG was plainly visible; by showing a peculiar pattern, it provided the clues to the diagnosis without requiring further analysis. Provocatively, Gibbs, Davis, and Lennox (1935, p. 1134) stated in their first paper on the EEG: "The method is exceedingly simple." Above all, the statement reflected their own perplexity as Lennox and Gibbs had devoted a long and frustrating decade to investigating epilepsy by other means. But their provocation incensed others. Their colleague Herbert Jasper reacted particularly strongly (Jasper, 1936, p. 1131): "I believe that it is important to point out that the method is actually exceedingly complicated." This was more than a quarrel about taste or character; it was a conflict between fundamentally different approaches. Jasper continued where Adrian had left the field and aimed to elucidate the neuronal mechanisms involved in the generation of brain waves. The goal was to integrate the EEG into the existing body of neurophysiological knowledge. Along this line of research, the EEG was a tool to visualize complex aspects of the neurons' communication code. Quite the contrary for Frederic and Erna Gibbs; for them, the EEG generated electrographic representations of diseases that could be read like images. Without further neurophysiological experimentation or mathematical analysis, the "trained eye of the electroencephalographer" should "see at a glance what it has taken others many hours to find," as they claimed in the preface to their seminal EEG atlas (Gibbs & Gibbs, 1941, p. VI):

For example, although it is possible to tell an Eskimo from an Indian by the mathematical relationship between certain body measurements, the trained eye can make a great variety of such measurements at a glance and can often arrive at a better differentiation than can be obtained from any single quantitative index or even from a group of indices. [. . .] A "seeing eye" which comes from complete familiarity with the material is the most valuable instrument which an electroencephalographer can possess.

The seeing eye required training precisely because the EEG generated representations of diseases in its own medium of the graphic record. In this approach, the visualization did not reveal generalizable neurophysiological knowledge but individual images that could be grouped and classified on a purely visual basis.

However, both approaches, visual thinking with the trained eye and experimental thinking with neurophysiological scrutiny, share a refusal to find the kind of meaning in the EEG that had initially fascinated public and scientific audiences alike. The fertility of the EEG as a diagnostic tool and its futility, by and large, in elucidating the brain's mode of operation contributed equally to extinguishing the early fascination and the hope that had been invested in it. By 1950, it was generally accepted that the "meaning" of the EEG was primarily clinical and its value for basic neurophysiological research limited. That does not mean that approaches to identify the cognitive or psychophysiological significance of the EEG were no longer pursued; they lived on, for example, in Wiener's cybernetic EEG studies, as discussed above. The difficulty in finding meaning in the EEG in cognitive terms resulted in enormous efforts to crack its code. Wiener mobilized MIT's recently acquired digital computing powers for this task (Barlow, 1997). Yet autocorrelation analysis was but one way to proceed; what about other frequencies or the relative distribution of rhythms in time or space? What about phase relations or periods of synchronization? There were ample opportunities for very different forms of analysis and visualization, as testified by the proliferation of new approaches to record, to display, and to analyze brain waves that Walter, Wiener's British counterpart, reeled off (Hayward, 2001). His laboratory in Bristol transformed into a zone of symbiosis or coevolution of human minds and machines (Walter, 1953, p. 193): "The machines that flesh and click in our laboratories now are the first forms of the living brain's extended life, the rudiments of racial understanding, as Gutenberg's first printing presses were the forerunners of the Reformation." Walter's tinkering with radar and television screens, flicker lamps, vacuum tubes, and frequency analyzers created an epistemological space that rotated less around the confirmation of a theory than around the productivity of a technological ensemble for envisioning the brain's new life.

Walter was the observer who had reported the Cambridge experiment for the *Spectator*, so succinctly concluding with: "So, with this technique, thought would seem to be a negative sort of thing." He spent his life proving the contrary. His EEG research continued to generate new theories as human brains continued to send encrypted signals worth every new form of visualization. But the solution that had seemed, in 1934, to be so close has long since vanished in the information that has amassed; on the whole, EEG research has made the picture much more complicated. If this is exemplary for the productivity of visualization in the neurosciences, then the manifesto by the German neuroscientists is right and wrong at the same time. New visualization techniques such as functional neuroimaging will change the picture rather than solve the riddle of mind and brain.

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